IBM Db2Hack 2026 — Round 1 Proposal

Db2 Sentinel AI: Autonomous Financial Crime Surveillance, Multi-Agent AML & Regulatory SAR Intelligence Platform

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1. Problem Statement

Global financial institutions face an estimated **$3.1 Trillion** in illicit money laundering and terrorist financing annually. While enterprise banking universally relies on **IBM Db2** for mission-critical core ledgers and high-throughput transactions, existing Anti-Money Laundering (AML) compliance tooling relies on brittle keyword rules and isolated point-solutions. This creates three critical failures:

• **95% False-Positive Rate:** Compliance teams waste 80% of operational hours reviewing benign transactions.
• **Investigation Fatigue:** A single alert takes 4 to 6 hours of manual ledger querying, counterparty tracing, and document drafting.
• **Severe Regulatory Fines:** Missing statutory deadlines (FinCEN 30-day mandate, RBI STR guidelines) incurs billions in fines.

Crucially, existing AI attempts extract sensitive financial data into third-party vector databases, causing serious data privacy breaches (GDPR, PCI-DSS).

2. Proposed Solution & Key Features

Db2 Sentinel AI transforms IBM Db2 into an active, autonomous regulatory intelligence hub. It integrates **IBM Db2 12.1+ Native Vector Search**, **Recursive Common Table Expressions (CTEs)**, and **LangGraph cyclical multi-agent orchestration** to autonomously detect financial crime and compile legally compliant Suspicious Activity Reports (SARs) in **under 30 seconds**.

- **In-Database Semantic Vector Matching:** Indexes FATF typologies using native Db2 VECTOR(384, FLOAT32) and VECTOR_DISTANCE(..., 'COSINE') within atomic transactions.
- **Recursive CTE Circular Flow Traversal:** Detects multi-hop circular fund routing (A → B → C → A) directly within IBM Db2 via SQL Common Table Expressions without external graph databases.
- **Window Velocity Analytics:** Employs Db2 window functions (AVG OVER PARTITION) to identify burst structuring patterns just below statutory reporting limits ($9,000–$9,999).
- **LangGraph Checkpoint Persistence in Db2:** Preserves intermediate agent states, tool outputs, and decisions in Db2 AGENT_CHECKPOINTS for ACID reliability, pause-and-resume workflows, and courtroom replayability.
- **Automated FinCEN / RBI SAR Filing:** Generates formal 5-part regulatory Suspicious Activity Reports with line-item citations to Db2 transaction record IDs.
- **Immutable Cryptographic Audit Trail:** Computes SHA-256 cryptographic signatures on all enforcement actions and stores them in the Db2 AUDIT_TRAIL table.
- **Dual-Mode Engine:** Operates natively on live IBM Db2 12.1+ (via ibm_db and SQLAlchemy ibm_db_sa), with an embedded zero-setup local emulation engine for instant reproducible evaluation.

3. Selected Agentic AI Framework

Framework: LangGraph (Cyclical Multi-Agent StateGraph) & **LangChain Core**.
Orchestration Pipeline:
1. **ScreeningAgent:** High-throughput rule & sanctions watchlist screening.
2. **ForensicInvestigatorAgent:** Queries historical Db2 ledgers and performs native Db2 Vector similarity search.
3. **GraphRiskAgent:** Recursive SQL CTE cycle discovery and network topology analysis.
4. **SARDrafterAgent:** Auto-generates structured regulatory compliance filings.
5. **SupervisorAgent:** Human-in-the-loop review, account freeze authorization, and cryptographic audit signing.

4. IBM Db2 Connector & Intended Integration

Drivers & Connectors: ibm_db (native C-driver bindings) and ibm_db_sa (SQLAlchemy dialect).
Db2 Capabilities Leveraged: Native VECTOR(384, FLOAT32) column type; VECTOR_DISTANCE(..., 'COSINE'); WITH RECURSIVE Common Table Expressions; analytical window functions (AVG OVER PARTITION); transactional ACID locking; and temporal audit logging.

5. Proposed System Architecture & Workflow

Workflow: Inbound Wire Interception → ScreeningAgent (Db2 Watchlist & Threshold Check) → ForensicInvestigatorAgent (Db2 Vector Typology Matching) → GraphRiskAgent (Db2 Recursive CTE Loop Detection) → SARDrafterAgent (FinCEN/RBI Regulatory Narrative) → SupervisorAgent (Human Gatekeeper & Db2 Freeze) → **IBM Db2 12.1+ Ledger** (Checkpoints, Audit & SAR Storage).

6. Expected Impact & Enterprise Value

• **82% Reduction in False Positives:** Vector-grounded typology matching distinguishes routine commercial transfers from genuine structuring.
• **90% Faster SAR Turnaround:** Generates complete, evidence-backed regulatory dossiers in under 30 seconds versus 4–6 hours of manual drafting.
• **Zero Data Leakage:** Sensitive financial transactions and vectors remain inside IBM Db2's enterprise security boundary.
• **Courtroom-Grade Auditability:** Every reasoning step and enforcement decision is immutably timestamped and cryptographically signed in Db2.